EESHITVAA SINGH

Multimodal Project

MUSIC

MUSIC

Secondary Research

Music (Categorization of Information)

INFORMATION PILLAR	DESCRIPTION & CONTEXTUAL ATTRIBUTES
1. Genre	The stylistic category (e.g., Classical, Jazz, Electronic) that sets user expectations.
2. Sound Type	The fundamental origin of the audio, ranging from purely acoustic to entirely synthetic.
3. Instruments	The physical or digital tools utilized (Strings, Percussion, Synthesizers, Controllers).
4. Culture	Geographical and historical context (e.g., Western tonal systems vs. Eastern microtonal).
5. Purpose	The intended function of the piece (Entertainment, Ritualistic, Focus/Flow state, Healing).
6. Properties	The core physical metrics of the sound wave itself (Tempo, Pitch, Timbre, Amplitude).
7. Vocal	The presence and style of human voice (A Cappella, Choir backing, Lead melodic vocal).

MUSIC

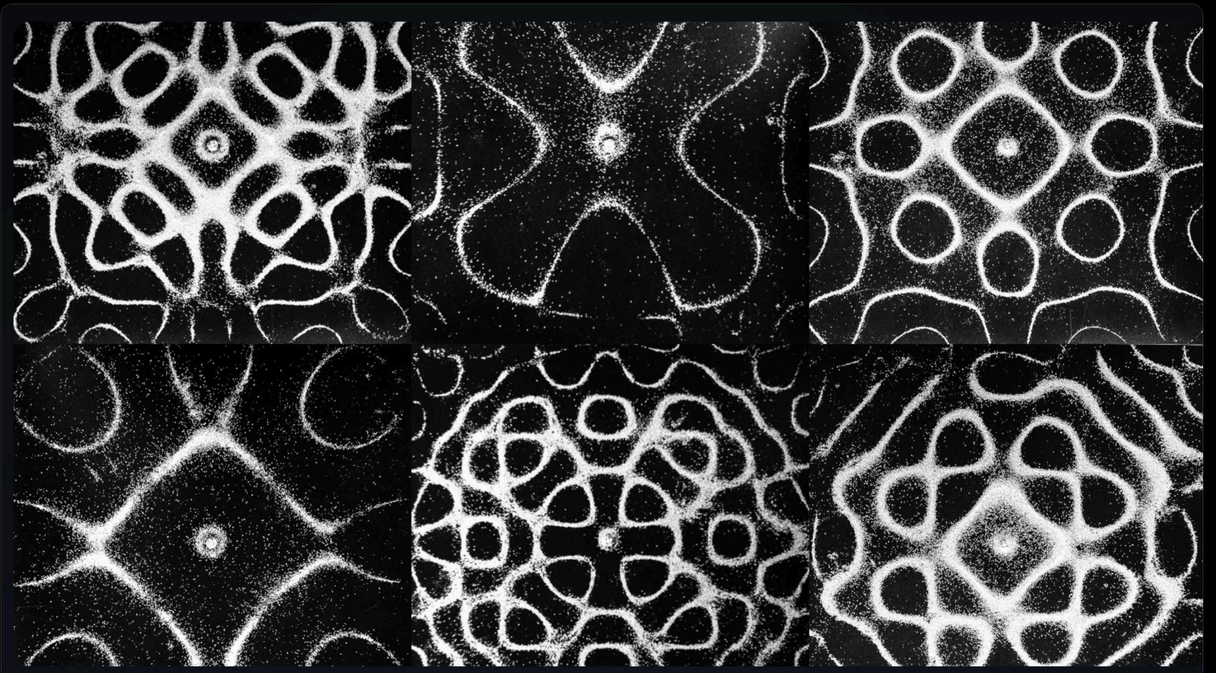
(CATEGORIZATION OF INFORMATION)

Genre	Sound Type	Instruments	Culture Origin	Purpose	Properties	Vocal
1. **Classical:** The Architecture of Information • 1000 CE – Present; Europe. • Musical Notation. By writing music down, the composer's brain was freed from the limits of memory. This allowed for Polyphony (multiple melodies at once). • It evolved to demonstrate Order, Power, and Logic. Baroque (1600-1750): Math-based (Bach). It sounds like a complex clock. Romantic (1820-1910): Emotion-based (Beethoven/Wagner). It uses huge orchestras to overwhelm the human nervous system. Modern: Breaking the rules to reflect a fragmented, industrial world. **2. Jazz:** The Democracy of Sound • The "When/Where": 1890s – 1920s; New Orleans, USA. • The "How": Syncopation (accenting the "weak" beats) and the Blue Note (a note slightly "out of tune" that mimics a human cry). It is built on "Call and Response." • The "Why": Jazz evolved as a Survival Language. Enslaved people and their descendants combined African rhythmic complexity with European harmonic instruments (Trumpet/Piano). It celebrates the individual through the solo. **3. Rock:** The Electricity of Rebellion • The "When/Where": 1950s; USA and UK. • The "How": The Backbeat (A heavy "thwack" on beats 2 and 4) and Amplification. The electric guitar turned a musical instrument into a weapon of volume. • The "Why": It evolved to serve the Teenage Demographic. After WWII, youth had money and a voice. Rock provided a "high-energy" sound that separated them from their parents' generation. 4. Pop: The Science of the "Hook" • The "When/Where": 1950s – Present; Global (centered in USA/UK/Sweden). • The "How": Averaging. Pop music uses the most common chord progressions (like I-V-vi-IV) because the brain finds them easy to process. It uses "Production Polish" (Auto-Tune, compression) to make the sound perfectly "clean." • The "Why": To achieve Mass Social Bonding. Pop is designed to be a "social lubricant"—music that everyone can agree on at the same time to trigger collective dopamine. 5. Hip-Hop/Rap: The Post-Industrial Collage • The "When/Where": 1970s; The Bronx, New York City. • The "How": The Breakbeat and Sampling. DJs realized they could take the "drum break" of an old record and loop it forever. This turned the Turntable into an instrument. • The "Why": It evolved as Rhythmic Journalism. In poor urban areas where instruments were expensive, the "Voice" and the "Record Player" became the tools to tell the story of the streets. 6. Electronic/EDM: The Digital Trance • The "When/Where": 1980s; Detroit (Techno) and Chicago (House). • The "How": The Four-on-the-Floor beat (a kick drum on every beat) and Synthesizers. It removes the "human error" of a live drummer to create a perfect, machine-like pulse. • The "Why": To induce Ecstatic States. Much like ancient tribal drumming, the repetitive 120-130 BPM (Beats Per Minute) matches the "arousal" heart rate of a dancing human, leading to a loss of "Self." 7. Folk & Country: The Living Archive • The "When/Where": Ancient – Present; Rural regions (Appalachia, Ireland, etc.). • The "How": Acoustic Portability. Instruments	1. **Acoustic Sound:** The Physics of Friction Acoustic sound is the result of Mechanical Energy moving through a physical medium (wood, metal, skin, or air). • An external force (a finger plucking, a hammer hitting, or lungs blowing) introduces energy into a material. This material has mass and elasticity, causing it to vibrate. • The Key Characteristic: Natural Decay. In the physical world, friction is king. As a string vibrates, it loses energy to the air and the bridge of the instrument. This causes the sound wave to "die" over time. • The shape of this decay is called the Envelope. Acoustic sounds usually have a sharp Attack (the hit) and a gradual Release (the fade). • The (The Human Connection): Our brains evolved over millions of years to listen specifically to acoustic sounds. We use the "imperfections" of acoustic sound (the scratch of a bow, the breath of a flute) to judge how far away a predator or a friend is. **2. Electronic Sound:** The Physics of Voltage Electronic sound removes the "physical" middleman. It is sound born from the movement of electrons in a circuit or code. • A Voltage-Controlled Oscillator (VCO) or a digital algorithm generates a steady, repeating electrical signal. This signal is then sent to a speaker, which turns the electricity back into air wiggles. • Infinite Sustain. Unlike a guitar string, a circuit does not get tired. It does not have physical friction. As long as the electricity is "On," the note stays perfectly flat and perfectly loud forever. • Electronic sound allows for "Pure Waves" (Sine, Square, Sawtooth). These waves are mathematically perfect and do not exist in nature. A Sine wave has zero overtones—it is the "atom" of sound. • It allows humans to "play the impossible." We can create a bass note so low it vibrates our internal organs, or a lead sound that changes its "flavor" (timbre) every millisecond—something a wooden violin could never do. **3. Ambient/Environmental:** The Physics of Space This category is not about a "tune"; it is about the Acoustic Signature of a location. Scientists call this Soundscape Ecology. • It is a "Stochastic" (random) collection of sounds. • Biophony: Sounds made by living organisms (birds, insects). • Geophony: Sounds made by the earth (wind, rain, thunder). • Anthropophony: Sounds made by humans (traffic, humming refrigerators). • The Key Characteristic: Non-Linearity. There is no "beginning, middle, or end" to a rainstorm. It is a texture. Ambient sound doesn't demand your attention; it surrounds it. Humans have an "Auditory Horizon." We use background sound to feel safe. If the birds in a forest suddenly stop singing (Biophony), the human brain immediately enters "High Alert" (anxiety). Ambient music (like Brian Eno's) replicates this feeling of safety by providing a predictable, calm environment. **4. White/Colored Noise:** The Physics of Probability "Noise" in physics is a signal that contains a random mixture of all frequencies. We use "colors" to describe how those frequencies are balanced. • White Noise (The "TV Hiss"): • Physics: It contains every frequency the	1. **Chordophones:** The Physics of Tension • Sound is produced by one or more strings stretched between fixed points. • When you pluck, bow, or strike a string, you create a Standing Wave. The frequency (pitch) is determined by the Law of Strings: a. Length: Shorter strings vibrate faster (higher pitch). b. Tension: Tighter strings vibrate faster. c. Density/Mass: Thicker strings vibrate slower (lower pitch). • A string by itself is almost silent because it is too thin to move much air. Chordophones require a Resonator (the wooden body of a guitar or the soundboard of a piano) to amplify the vibration through Acoustic Coupling. • Sub-Types: Lutes (with necks, like violins), Zithers (strings over a body, like pianos), Harps, and Lyres. **2. Aerophones:** The Physics of Air Columns • The instrument doesn't vibrate; the body of air inside it vibrates. • The Science: You introduce a "disturbance" at one end (blowing over a hole, vibrating a reed, or buzzing your lips). The tube acts as a Filter. It only allows sound waves that "fit" the length of the tube to survive. • The Secondary Research Detail: This is based on Bernoulli's Principle and fluid dynamics. By opening or closing holes, you are effectively changing the "Length" of the instrument, which changes which frequency can resonate inside. • Sub-Types: Flutes (edge-blown), Reed instruments (oboe/sax), Brass (lip-vibrated), and Free aerophones (like a siren or a bullroarer). 3. Membranophones: The Physics of 2D Surfaces • The "How" (Mechanism): Sound is produced by a tightly stretched membrane (skin or plastic) vibrating. • The Science: Unlike strings (which are 1D lines), membranes are 2D planes. This creates much more complex vibration patterns called Modes. • The Secondary Research Detail: Most membranophones produce inharmonic Overtones. This is why a drum usually sounds like a "thud" or a "crack" rather than a "singing note" like a flute. To get a "tuned" drum (like a Timpani), the shape of the kettle must be mathematically precise to "force" the membrane into harmonic patterns. • Sub-Types: Struck drums, Plucked drums, and Friction drums. 4. Idiophones: The Physics of Material Elasticity • The "How" (Mechanism): The whole substance of the instrument vibrates. No strings, no membranes, no air. • The Science: These instruments rely on the Inherent Elasticity of the material (metal, stone, wood). When you hit a gong, the metal atoms compress and expand, creating a wave through the solid material that then pushes the air. • The Secondary Research Detail: This is the most ancient category (40,000+ years old). It is also the most "efficient" because the resonator and the vibrator are the same object. • Sub-Types: Struck (Cymbals), Plucked (Jew's harp), Friction (Glass harmonica), and Shaken (Maracas).	**Ancient caves** (like those in Lascaux or Chavin de Huantar) were chosen because they had specific echoes that mimicked the human voice or a heartbeat. "Music" didn't start with a string; it started with a man humming in a cave until the cave "hummed back." • Archaeological evidence, such as bone flutes found in European caves, suggests music existed at least 40,000 years ago, though it likely existed in more basic vocal forms long before(Slovenia and Germany) • Mimicking Nature: Early humans probably imitated bird songs and other natural sounds for communication, hunting, and warnings. • Vocal & Rhythmic Roots: Music began through humming, shouting, and singing, coupled with rhythms from clapping or stamping, creating social bonds. • Ancient Instruments: Early instruments included flutes made from vulture bones or mammoth ivory, drums, and rattles. **1. Western:** The Architecture of Harmony Western music (Europe/North America) is unique because it evolved to be Vertical. While most of the world focused on a single beautiful melody, the West focused on "stacking" different notes on top of each other. • The "How" (The 12-Tone Scale): To make Harmony work, Westerners had to "standardize" pitch. They divided the octave into 12 equal parts (Equal Temperament). • This was a mathematical compromise. By making every note slightly "out of tune" with the laws of physics, they allowed instruments to play in any key. This led to the invention of the Piano and the Symphony Orchestra. • The "Why" (The Church & Notation): The Catholic Church needed thousands of monks to sing the same thing. They invented Staff Notation (writing music down). Once music was on paper, composers could "design" massive structures like a skyscraper. • Research Focus: Verticality. It is about the "Chord"—multiple voices moving together in a logical, mathematical progression. **2. Eastern/Asian:** The Melodic Continuum In Eastern traditions (specifically India, China, and Japan), music is Horizontal. It focuses on the infinite "shades" of a single note rather than jumping between chords. • While a Western piano has 12 notes, Indian Classical music recognizes 22 or more "microtones" (Srutis) in an octave. • Secondary Research Detail: Musicians use Meend (gliding) to slide between notes. If Western music is like stairs (step by step), Eastern music is like a slide (continuous movement). • Raga System: A Raga is more than a scale; it is a "personality." There are Ragas for the morning, Ragas for the monsoon, and Ragas for midnight. • In the Vedic tradition of Ragas, certain melodies were forbidden to be played at the wrong time. A morning Raag played at night was considered "poisonous" to the nervous system. 3. African: The Polyrhythmic Foundation African music (Sub-Saharan) is the "Engine" of modern music. It treats Time as a three-dimensional object. • The "How" (Polyrhythm): In Western music, everyone usually follows one "beat." In African music, different instruments play different "clocks" at the same time (e.g., one person plays in 3, another in 4). • Secondary Research Detail: This is built around a "Timeline" (often played on a bell). It is a repetitive cycle that never	1. **Performance:** The Cult of the Virtuoso • Music designed to push the limits of human physical capability. It relies on the Concert Model: a silent, passive audience focusing all attention on an active, "super-human" performer. • To demonstrate Biological Fitness. In evolutionary terms, a virtuoso (like Paganini on violin or Jimi Hendrix on guitar) is signaling high intelligence, fine motor coordination, and discipline. • This creates a Parasocial Interaction. The audience "bonds" with the performer's ego. It is about the "Awe" response—the brain's reaction to witnessing something that seems impossible. **2. Film & TV Scores:** The Emotional Compass • Using Leitmotifs (musical "tags" for characters or ideas). When the "Vader" theme plays, the brain knows "Evil" is present before the eyes see it. • To resolve Visual Ambiguity. The human brain is highly visual, but eyes can be tricked. Music tells the brain how to interpret the image. • Example: A scene of a man walking into a house is neutral. Add minor-key strings, and it's a horror movie. Add upbeat jazz, and it's a comedy. • This is based on Affective Priming. The music "primes" the amygdala to feel a specific emotion, making the visual story feel more "real." **3. Religious & Ritual:** Marking Sacred Time • Using repetitive, predictable structures (chants, hymns, bells). It often uses Acoustic Grandeur (huge reverb in cathedrals or loud drums in open fields) to make the individual feel "small." • To create Collective Effervescence (a term by sociologist Émile Durkheim). It blurs the line between "Me" and "The Group." • Ritual music marks the transition from Profane Time (daily life) to Sacred Time (weddings, funerals, prayer). It acts as a "Social Clock" that tells everyone in a community to feel the same thing at the same time. **4. Dance:** The Motor System Hack • Using Neural Entrainment. High-energy, repetitive beats (120–140 BPM) that target the Basal Ganglia(the part of the brain that controls movement). • The Purpose: To foster Social Synchronization. When a group moves in perfect rhythm, they release Oxytocinand Endorphins. This creates a "Warrior Bond" or "Tribal Trust" without needing to speak. • Dance music uses Syncopation (missing beats) to create "Groove." The brain wants to "fill in" the missing beat with a physical movement (a clap or a step), making the dance involuntary. **5. Therapeutic:** The Biological Repair Kit • Using music to target specific brain injuries or mental states. • Melodic Intonation Therapy (MIT): Helps people with Aphasia (who can't speak) to sing their words. Speech and Singing use different sides of the brain; the music "bypasses" the damage. • To achieve Homeostasis (balance). Music can lower Cortisol (stress) and increase Serotonin. • Neurochemical Release: Listening to or making music stimulates the release of dopamine (pleasure), serotonin (mood regulation), and endorphins (pain relief). • Stress Reduction: It effectively lowers cortisol levels, the body's primary	1. **Tone vs. Noise:** The Geometry of the Wave In physics, the difference between a "musical note" and "static" is purely a matter of Geometry and Predictability. • Tone (Periodic Waves): • The "How" (Mechanism): A vibration that repeats at a constant rate. Because the wave is symmetrical, the brain can "count" the cycles per second (Frequency). • The Research Detail: This is based on Fourier's Theorem. It states that any complex tone is actually a "stack" of simple sine waves. When these waves are in a mathematical ratio (1:2, 2:3), we hear a "Clear Pitch" that we can hum or tune. • The "Why" (Human Response): The brain loves tones because they are Predictable. Predictability lowers cognitive load, allowing us to perceive "Harmony." • Noise (Aperiodic Waves): • The "How" (Mechanism): A vibration with no repeating pattern. It is a "Chaotic Signal" where frequencies clash at random. • The Research Detail: "Noise" in music (like a snare drum or cymbals) provides Texture. Without noise, music would sound "clinical" or "dead." • The "Why" (Human Response): Noise is used for Auditory Masking. Because noise contains so many random frequencies, it "covers up" distracting sounds, which is why we use it for sleep and focus. 2. Infrasound: The "Fear Frequency" (< 20 Hz) Infrasound is sound that is too "slow" for the human ear to turn into an electrical signal, but the body still feels the physical pressure. • The "How" (Mechanism): These waves are massive (some are longer than a house). Instead of vibrating the eardrum, they vibrate the Large Organs and the Chest Cavity. • The "Ghost" Research. In a famous study by Vic Tandy, it was found that a frequency of 18.9 Hz caused the human eyeball to vibrate. This creates "visual smearing" or shadow figures in the peripheral vision. • The "Why" (Evolutionary Response): In nature, infrasound is created by earthquakes, avalanches, and tigers (their roar has an infrasound component). • Secondary Research Detail: Our brains associate these low-frequency "rumbles" with Imminent Danger. Film directors use "sub-bass" infrasound to create a feeling of "dread" in horror movies without the audience knowing why they are scared. 3. Audible Sound: The Human Window (20 Hz – 20,000 Hz) This is the biological "Interface" between the universe and the human mind. • The "How" (The Biological Filter): Inside your ear, the Cochlea is lined with tiny hairs. The hairs at the very beginning respond to high sounds; the ones at the very end respond to low sounds. • The Research Detail: Presbycusis. High-frequency hair cells are the most fragile. As we age, or are exposed to loud noise, these cells die and never grow back. • The "Mosquito" Effect: Teenagers can hear a high-pitched "ring" (around 17,000 Hz) that adults over 30 physically cannot hear. This is used in "Teen Repellents" (devices that play a sound only kids can hear to stop them from loitering). • The "Why" (Evolutionary Sweet Spot): Human speech usually sits between 500 Hz and 4,000 Hz. Our ears are physically "tuned" to be most sensitive in this range so we can understand language in a noisy environment.	1. **Vocal Music:** The Biological Signature The human voice is the only "Internal" instrument. It is the only sound source that is physically part of the performer's body. • It is a three-part biological engine: a. The Power (Lungs): Controlled air pressure. b. The Vibrator (Larynx): Two flaps of muscle (Vocal Folds) that chop the air into pulses. c. The Resonator (Vocal Tract): The throat, mouth, and nasal cavities. • Humans use Formant Tuning. By changing the shape of our mouths, we create specific "peaks" of energy in the sound wave that allow us to project over an entire orchestra without a microphone. • The voice carries the "Signature of the Self." Through a voice, the brain instantly detects: • Emotion: Fear, joy, or sadness (via the Limbic System). • Identity: Gender, age, and size. • Intention: Is the person threatening me or soothing me? • The Power of Language: Because vocal music usually uses words, it engages the Left Hemisphere (Wernicke's Area) for logic and meaning. It creates a "Specific Reality"—if the singer says "I love you," the meaning is fixed. 2. Instrumental Music: The Abstract Tool Instrumental music is the "Externalization" of the human soul. It uses tools to produce sounds the body cannot physically make (e.g., a note held for 10 minutes or a sound louder than a jet engine). • The "How" (Mechanism): It is an Extension of the Body. Whether it is a piano or a flute, the instrument acts as a "Prosthetic" for the human hand or lung. • Secondary Research Detail: Instrumental music relies on Timbre (Tone Color). While all voices sound like "humans," instruments can sound like "trees" (woodwinds), "earth" (drums), or "stars" (synthesizers). • The "Why" (Narrative Freedom): Instrumental music is "Semantically Weak." Because there are no words, the listener is not "told" what to think. • Cognitive Result: The brain's Default Mode Network (the daydreaming center) is activated. You hear a violin and you imagine a landscape or a memory. The music is a "Mirror" for your own internal state. • The Universal Bridge: Instrumental music has no Language Barrier. A person in Tokyo and a person in Rio can listen to the same Bach Cello Suite and feel the same "Math and Beauty" without needing a translator. 3. The "Uncanny Valley": When One Mimics the Other Evolutionarily, these two types are constantly trying to switch places. • Instruments Mimicking Voice: Musicians spend years learning to make their instruments "Sing." • Example: The Violin and the Cello are shaped and tuned to mimic the frequency range of the human Soprano and Baritone. When a cellist uses "Vibrato," they are literally mimicking the "tremble" of a human voice in emotion. • Voice Mimicking Instruments: In genres like Jazz Scatting, Beatboxing, or Bulgarian Choir, the voice is stripped of "Language" and used as a pure "Aerophone." • Research Detail: When the voice loses words, the brain stops processing it in the language center and moves it to

Research Insights

1. Music isn’t only just what you hear, it forms pattern ,each note has a colour ..**when a specific note is touched or called for it forms a very specific and perfect gometry ...like a shape in the air** ,which not only lets us listen to music but also form an image and toneality of it .

2. Some sound feels like a noise while some melody because **when we hear a new sound we try to understand it like it’s a puzzel ...if we understand it or it seems familiar to us (something that we have heard many times)**..that’s why these days when we here something new on insta or social media we don’t like it immediately but the we hear it so many times that we start liking it ..



JAN 9, 2025

Chladni Patterns: A Visual Representation of Acoustic Wave Mechanics

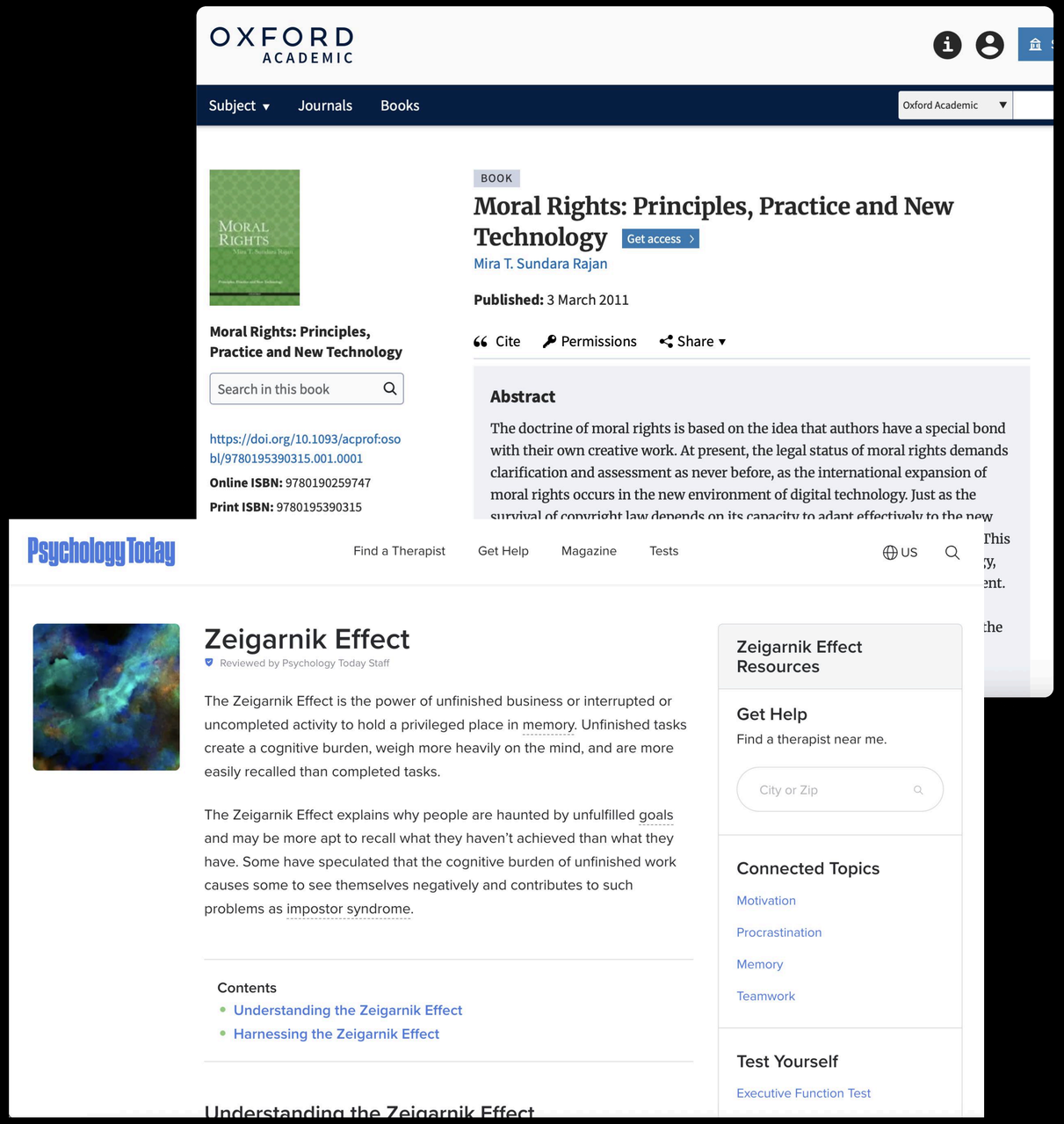
RESONIKS studies the intersection of acoustics and wave mechanics to perform Acoustic Resonance Testing, or ART. But did you know that you can use these same principles of wave mechanics to create actual art? This phenomenon, called Chladni patterns, where vibration creates geometric patterns, provides a rare glimpse into

3. Western music is "Vertical"—it's built on chords and progress. It makes us think about "what will happen next" (Future-oriented). Indian Classical music (Sitar) is "Horizontal"—it is built on a single drone. It forces the brain to stop looking for "progress" and start looking for "detail" within the same moment... that's why music in a way forms a perception of a particular culture .

4. When the Sitar plays over that C# drone, **our brain stops processing- thinking about what will happen next and starts experiencing**. This is why "Ambient" or "Drone" music is used in every spiritual tradition in history—it is a psychological "mute button" for the ego- to be in the present .

5. If a melody stays on the 7th note (the "Leading Tone") and never returns to the "Home" note , **our brain stays in a state of high cognitive load...to listen the last note ...**that's why sometimes when we listen to a song ..if we don't hear the end of that particualr lyrics and song ,**we feel an itch to listen it ...that's what great musicians also do in concerts and in clubs too.**

6. People from California hear certain pairs of notes as "ascending," while people from London hear the exact same notes as "descending." **Your brain has a "geographical map" of pitch.**



Competitive Analysis

Feature	Traditional Instruments (e.g., Sitar/ Piano)	Streaming Apps (Spotify/YouTube Music)	Wellness Apps (Calm/Headspace)	Functional AI Audio (Endel/ Brain.fm)
Learning Curve	Extreme (7–10 years for mastery)	None (Passive listening)	Low (Follow a guide)	None (Algorithm-led)
Tactile Feedback	High (Physical strings/vibration)	Zero (Digital glass screen)	Zero (Voice-led)	Low (Headphones only)
Cognitive Load	High (Requires intense focus)	High (Lyric fatigue/Choice paralysis)	Medium (Listening to instructions)	Low (Designed for background)
Creative Agency	Full (You create the sound)	None (Spectator)	None (Consumer)	None (Algorithm-led)
Atmosphere/ Soul	Organic/Ancient	Digital/Hollow	Generic/Synthetic	Mathematical/Cold
Time to "Peace"	Slow (After hours of practice)	Variable(Depends on skip-rate)	10-15 mins	5-10 mins
Visual Connection	None	None/Album Art	Calming videos	Dynamic UI

Primary Research

User Interviews

06
Professionals

05
Musicians

06
Students

02
Teachers

Experts Interviews

SEGREGATION OF INTERVIEWS

SEGREGATION OF INTERVIEWS

US1	US2	US3	US4	US5	US6	US7	US8	US9	US10	US11	US12	US13	US14
• Name: Munajit Vishwas | Age: 38+ | Location: Calcutta • Occupation: Professional (Context suggests a corporate/ office background in a metro city). • Lifestyle & Context: Living in the cultural hub of Calcutta, Munajit is a witness to the transition from "Original" music to the "Remix Era." He is a family-oriented man whose emotional world is anchored by his parents and child. • Musical Philosophy: He views music as a Situational Mirror. He doesn't believe in "fixed" favorites; instead, the music must match the specific life event or emotion he is navigating. He holds a firm belief in the "Old is Gold" mantra, viewing modern music as an expansion of existing ideas rather than new innovation. • Instrumental Journey: He grew up around a sister who played the Harmonium, leading him to the Tabla. However, the "Hustle" of adult life and a lack of mental concentration forced him to stop. • Deep Insight: For him, the most vital "sound" isn't a melody but the Human Name. He prioritizes relational sounds over artistic ones, suggesting a soul that seeks connection over entertainment.	• Name: Anjali Sharma | Age: 40 | Location: Banaras • Occupation: School Teacher (Manages students, syllabus, and grading). • Lifestyle & Context: Her daily life is a storm of "noise"—screaming students, honking rickshaws, and domestic chores. She is a woman seeking a "mental exit" from her high-pressure responsibilities. • Musical Philosophy: Music is her "Pause Button." It is a functional tool to tell her brain to stop working. She suffers from "Lyric Fatigue"—because she listens to words all day at work, songs with lyrics make her brain "keep working" when it should be resting. • Instrumental Journey: She is an admirer of the Piano, viewing it as a beautiful vibration that "stays in the bones." However, she feels "foolish" attempting to learn it now, viewing it as a giant, intimidating puzzle. • **Deep Insight:** She experiences "Atmospheric Frustration." She will spend 20 minutes looking for a song to match a rainy day, find nothing that feels "deep" enough, and choose to sit in total darkness instead.	• Name: Kriti | Age: ~20 | Location: Delhi • Occupation: Design Student (UX/UI focus). • Lifestyle & Context: A young student navigating the fast-paced life of Delhi. Her life is a mix of academic stress and the modern "Swiftie" fandom. • Musical Philosophy: Music is Instinctive & Reactive. It isn't a hobby she "attends" to; it is a spiritual response to being tired or worried. She balances high-energy Bollywood with the deep, relatable songwriting of Taylor Swift. • **Deep Insight:** She represents the "Instant-Gratification Listener." She doesn't need to "watch" music or see a video; she only needs the sound to trigger an immediate emotional shift.	• Name: Poonam | Age: 38 | Location: Prem Nagar • Occupation: Homemaker / Local Worker (Context suggests a community-oriented lifestyle). • Lifestyle & Context: A resident of a smaller locality, she finds comfort in the past. She views the modern world as "too pop" and lacking emotional depth. • Musical Philosophy: Music is an Auditory Sanctuary. She is a pure listener who rejects the visual or physical aspects (dancing/watching) of music, preferring the internal experience of old melodies. • Instrumental Journey: She had a deep desire to sing but suffered from "Internalized Discouragement." She believed she didn't have "the talent" and stepped back before she could even try, representing a massive demographic of people who quit due to low self-esteem. • Deep Insight: She feels modern music is "hollow," proving that for her, music must have a historical or nostalgic weight to be valid.	• Name: Muneesh | Age: 55 | Location: Saharanpur • Occupation: Likely a Retired Professional or Senior Artisan. • Lifestyle & Context: A man of deep roots who values "Hindi Culture" and slow living. His world is one of soft sounds and gentle transitions. • Musical Philosophy: Music is Quality of Sound. He is sensitive to decibels and the "loudness" of modern music. He seeks the "unique sound" of an instrument that can be played both loudly and gently. • Instrumental Journey: A self-taught Flutist who learned by Observational Legacy. He simply picked up his father's instrument. He views the flute as an extension of the breath, something "easy" if you have the soul for it. • Deep Insight: His "perfect sound" is Rain. This highlights a person who finds man-made music inferior to the natural rhythms of the earth.	• Name: Ameesha | Age: 20 | Location: Dehradun • Occupation: Full-time Employee (Works a desk/ service job). • Lifestyle & Context: A local girl working a long shift. By the time she gets home, she is exhausted, leaving her with a "heart-desire" for music but "physical-fatigue" that prevents her from practicing. • Musical Philosophy: Music is Emotional Translation. It is how she explains her feelings to herself when she is lonely or missing someone. • Instrumental Journey: She owns a guitar—a physical symbol of her dreams—but it sits unused because of "Time Poverty." • Deep Insight: She like music a lot and find peace in it which also made her buy guitar in the hopes of learning but she couldn't learn it becoz of lack of time and guidance	• Name: Gudpreet Singh | Age: 20 | Location: Amritsar • Occupation: Design Student. • Lifestyle & Context: A creative student who interprets the world through visuals and art. He is a "Fellow Swiftie" who uses music to cope with the absence of social connection. • Musical Philosophy: Music is Visual Gradient. He sees colors (Red, gradients) and "musical notes" in his mind when he hears sound. For him, music is a communication tool between the artist's soul and the listener's eye. • Instrumental Journey: A "YouTube Scholar." He learned the Kalimba because it looked "cool" in a Naruto video. He quit the guitar because it lacked the "soul" he was looking for and felt too disciplined. • Deep Insight: He uses music as "Emotional Fill." In a world where he might not have a person to talk to, Taylor Swift's lyrics act as his companion.	• Name: Harman | Age: 19 | Location: Delhi • Occupation: Animation & VFX Student (4th Semester). • Lifestyle & Context: An artist who draws and animates, his brain is wired to "dig into things." He is highly aware of geopolitics and social causes. • Musical Philosophy: Music is a Human Privilege. He views the ability to hear and feel music as a surreal gift that separates humans from animals. He believes music should be "Political and Heartfelt." • Instrumental Journey: A Percussionist. He leans toward the Drums because of the physical release. He learns through "Hands-on Experimentation" rather than a rulebook. • Deep Insight: He has "Auditory Hyper-Focus." He cannot use music as background noise for studying because he "respects" the music too much and focuses on every note, which ruins his concentration for other tasks.	• Name: Veeneda | Location: Dehradun • Occupation: Clinical Psychologist (UPES Mental Health Initiative). • Lifestyle & Context: A medical professional who views music as a pharmaceutical-grade tool for brain-wave regulation. • Musical Philosophy: Music is Biological Therapy. She analyzes the transition from Beta (stress) waves to Alpha/ Theta (relaxation) waves. She believes in "Slow, Classical" music over the "Noise" of modern high-decibel tracks. • Deep Insight: She warns against "Dopamine Anticipation." She believes streaming apps (Spotify/YouTube) make people addicted to "skipping" songs, which eventually kills the brain's ability to have deep relationships with anything.	• Name: Aakrit Karsha | Age: 20 | Location: Pune • Occupation: Product Design (PD) Student. • Lifestyle & Context: A high-achiever who has been playing music since age 3. He chose Design over Music as a career because of the "instability" of the music industry. • Musical Philosophy: Music is a Necessity & Experience. It is not an "enjoyment" but a core part of human existence. • Instrumental Journey: Plays Tabla, Harmonica, Guitar, Piano, and Drums. He prefers the guitar for its "freedom of space." • Deep Insight: He rejects "Convenience Technology." He believes that for music to be "soulful," it must hurt. He values the cut of the strings on his fingers, calling it "Sweet Pain."	• Name: Shriyanshi | Age: 20+ | Location: Agra • Occupation: Aerospace Engineering Student (B.Tech). • Lifestyle & Context: A high-level intellectual who balances the "Mathematical Relations" of Aerospace Engineering with the "Sangeet Prabhakar" degree in Classical music. • Musical Philosophy: Music is Structured Energy. She views Indian Classical music as a disciplined "Art" that connects to seasons (Raag Bahar for Rain). • Instrumental Journey: An expert in Piano, Guitar, and Tabla. She started at age 3 and attended a specialized music school instead of a normal school early on. • Deep Insight: She critiques the "Lack of Consistency" in modern learners. She believes people fail because they want to play a "song" in two days rather than respecting the "boring" foundation of theory. To her, music is a discipline, not a hobby.	Occupation: Software Engineer (High cognitive load, 10+ hours of screen time). • Lifestyle & Context: Lives in a high-traffic city (Bangalore). His commute is his primary "Transitional Space," but it's currently being ruined by "Skip-Fatigue." • Musical Philosophy: Music is "Survival Gear." It is a protective wall meant to block out the "Static" of his crowded mind. • The Creation Gap: He represents the "Dormant Creator." He owns the tech (MIDI keyboard) but lacks the "Brain bandwidth" to learn. He is a prime candidate for a "Short-cut to Flow State." • Deep Insight: He describes his brain as "100 Chrome Tabs Open." This is a perfect metaphor for modern mental crowding. He seeks "Deep Neon Blue" (Peace) to overwrite the "Jagged Red" (Stress).	A cerebral Design Professor from Dehradun who transitioned from "passive listener" to "active creator" by picking up the Sitar in 2017. His brain is wired for structure and aesthetics, leading him to seek soothing frequencies and narrative depth over loud noise. His musical soul is rooted in childhood memories of his sister playing Ghazals. • Wants & Needs: • A "Self-Anchor": Needs music as a stabilizing force to build a deeper relationship with himself. • The "Visual Soul": Desires multi-sensory immersion (like Pink Floyd) to bridge the gap between sound and vision. • The "Zone": Seeks technology that helps unlock the "universal power" of seeing colors and gradients in sound. • Cumulative Complexity: Prefers learning that builds slowly, allowing the brain to overcome the fear of "tricky" tasks. • Frustrations: • The Mastery Myth: Rejects the "Morning/ Noon/Evening" fallacy where beginners expect instant concert-level results. • Mechanical Boredom: Struggles with the "boring" foundation of repetitive scales, which acts as a filter for most learners. • Physical Strain: Drained by the "beating" his fingers and back take as an adult learner. • Digital Disconnect: Feels modern music lacks the immersive visual soul of the "Old Masters." • Key Insight: Amit identifies the "Instant Mastery Myth" as the primary reason 90% of beginners quit. He believes music must be a lifelong "Self-Anchor" rather than a quick fix for loneliness.	A 28-year-old musician living in the serene mountains of Uttarakhand. An adult learner (6 years) who chose music autonomously to express what was "already inside." His practice is non-compulsory but deeply regular, guided by his internal headspace rather than a clock. • Wants & Needs: • Mind-Body Bridge: Needs music as a tool for biological well-being and self-expression when words fail. • The "Child-State": Seeks to bypass "Cognitive Load" (thoughts of money/ work) to reach pure learning. • Acoustic Purity: Values the ear-to-hand connection; prefers tuning 18 strings by ear over using electronic tuners. • Exploration of Loneliness: Uses music to feel "good within himself" rather than as a quick fix for social isolation. • Frustrations & Challenges: • The "Quality" Barrier: Believes 90% quit because they buy bad instruments. When a tool lacks soul, beginners blame themselves and lose interest. • Physical Adaptation: Drained by the "sweet pain" of cuts, strained posture, and the years required to "rub the skill" into the body. • Digital Hollowness: Finds apps like GarageBand "hollow"; believes a digital interface cannot replace the evolving relationship with a physical instrument. • Key Insight: Utpal identifies "Cognitive Load" as the primary barrier for adults. Unlike children, adults cannot easily quiet the "mental noise" of life to enter a flow state. He also stresses that high-quality tools are more vital than lessons for preventing early burnout.

INSIGHTS

1. While users seek music to find "Shanti" (peace) and regulate their nervous systems , **modern technology (streaming/skipping) has turned listening into a high-speed hunt for a "dopamine hit."** This creates "Choice Paralysis." People spend 20 minutes looking for a song only to sit in silence because the digital delivery of music feels too "fast" or "hollow" to match a deep emotion they are feeling .

2. People really want to make music, not just listen to it. They want that "peaceful, in-the-zone" feeling that experts get when they play but most people work all day (9-to-5) and are too tired or busy to practice and also it takes 7-8 years to get good at an instrument like the piano or flute.

3. Users feel that modern digital music is "hollow" because it lacks the resistance of a physical instrument. Any digital tool designed for these users needs to mimic the "tactile struggle" to feel authentic.

4. People in loud, busy cities use music to create a **"quiet bubble" to escape the constant noise (traffic, crowds, honking) that tires them out.The ride to and from work is the most important time of the day. It's the only time they can get mental peace before getting home.** Music isn't just for fun anymore—it is a shield.

Stakeholder Mapping

STAKEHOLDER LIST

PRIMARY

Cultural & Music Patrons

1. Sonic Explorers/Music Enthusiasts
2. Learners

The Creative Visionaries

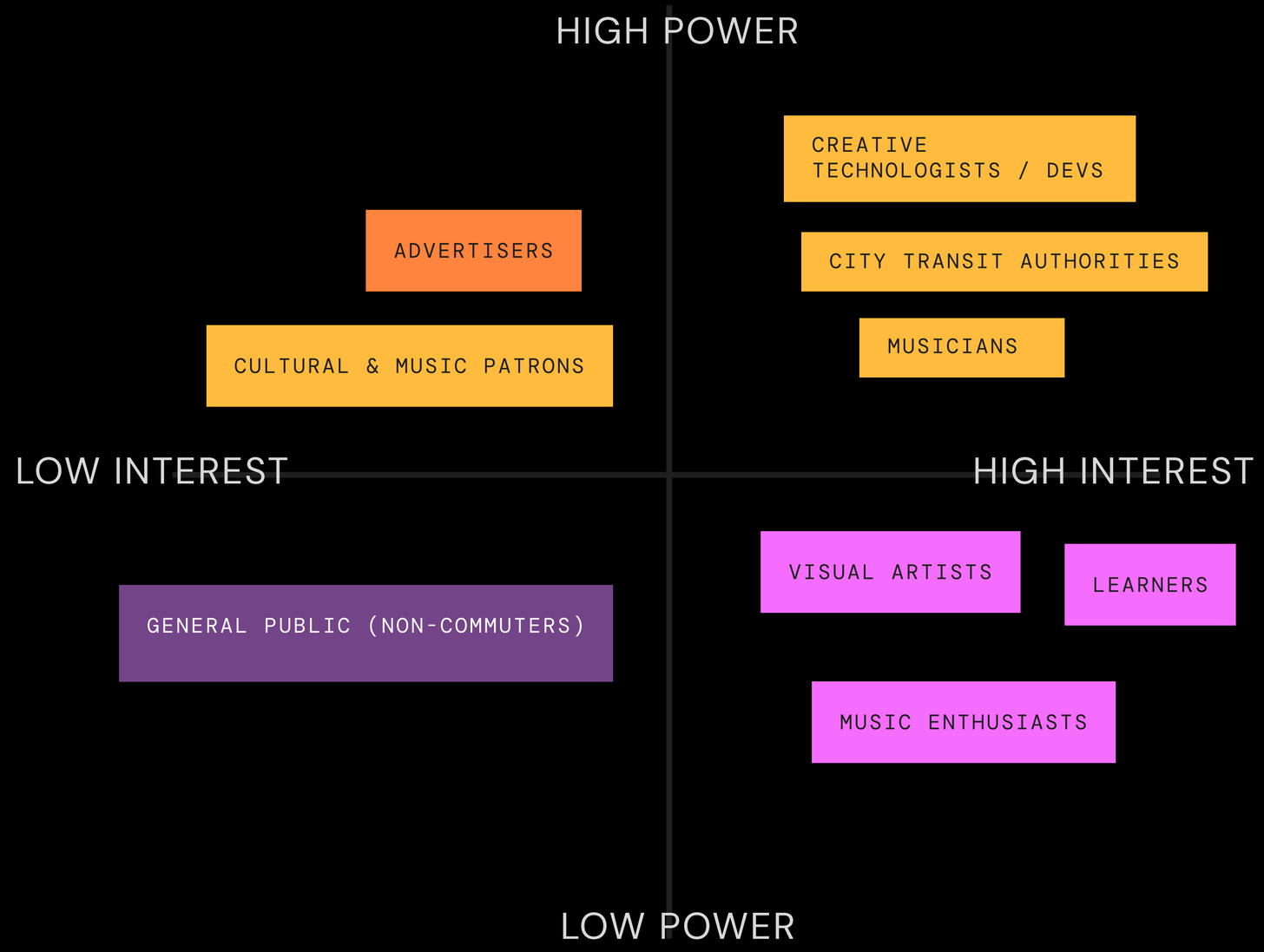
1. Visual Artists
2. Creative Technologist / Developer
3. Sound engeneers

Health & Wellness

1. Visually impared people
2. Doctors - Psychologist

SECONDARY

1. Musicians



5 Whys

1. Commuters arrive at their destination physically and mentally exhausted.

▼ (WHY?)

2. The daily travel environment is a "Sensory Battlefield" of honking, crowds, and mechanical noise.

▼ (WHY?)

3. This constant urban "Static" keeps the nervous system in a state of high anxiety and hyper-vigilance.

▼ (WHY?)

4. To survive, people use music as a "Shield" to escape reality and numb themselves to their surroundings.

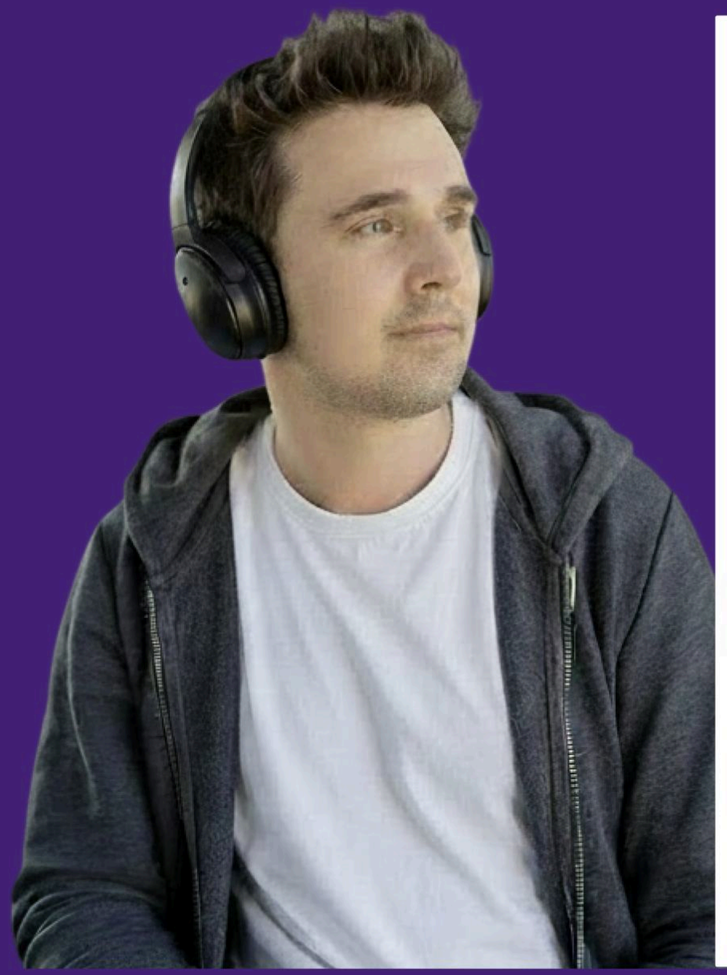
▼ (WHY?)

ROOT CAUSE

Urban infrastructure is designed for "Moving Bodies" but fails to "Feed the Soul," turning the human commute into a period of sensory survival rather than a moment of life.

User Persona

For Daily Commuters



Rohan Mehra

- 24 year old
- New Delhi / Dehradun (Urban)
- Junior software engineer
- Single , living in a shared apartment

Bio

Rohan, a 24-year-old Delhi developer, finishes work with a brain feeling like “100 open Chrome tabs.” His noisy, hour-long commute makes music vital “survival gear” to block out the city. However, he is increasingly drained by the mental effort required to find the right tracks to actually decompress.

Wants & Needs

- Wants to reach a "thoughtless" state of mind within seconds of putting on headphones.
- Needs a tool that creates a "private bubble" to block out the 80% of ambient city noise.
- Needs soulful, "thick" sounds without lyrics so his linguistic brain (burnt out from coding/ reading) can rest.
- Wants to learn music but don’t get the time

Tech & Lifestyle

Internet Usage:	●●●●●
Social Media:	●●●○○
Streaming Apps:	●●●●●
Gadget Enthusiast:	●●○○○
Skip-Rate (Vibe Search):	●●●●●

Pain Points

- **Skip-Fatigue:** Spends half his commute clicking "Next" because the algorithm misses his specific vibe.
- **Lyric Load:** Lyrics feel like "extra data" to a brain already at maximum mental capacity.
- wants to learn an instrument but can’t
- **Urban Noise:** Drained by constant honking and chatter that prevents a peaceful transition home.
- Even after listening to songs still feeling tired after coming back to home .

User Persona

For Musicians

Natsu Das

21
Agra (Urban Intellectual)
Aerospace Student / Sangeet Prabhakar (Vocal & Piano) Student

Tech & Lifestyle

Internet (Research/Theory) ●●●●●
Social Media (Fast Content) ●○○○○
Avoids "noisy" trends
Production Software (DAWs) ●●●●○
Tactile Instruments ●●●●●
Consistency/Discipline ●●●●●

Bio

Her life is a balance of extreme logic and extreme soul. He attended a specialized music school at age 3 before he even started traditional schooling. Today, she balances the complex calculations of Aerospace Engineering with the structural rigors of Indian Classical Music.

Wants & Needs

- Needs the physical resistance of strings or keys; he believes that without "tactile conviction," music is hollow.
- Needs to achieve that rare state where the external world disappears and 4 hours of practice feel like 10 minutes.
- Desires soundscapes that follow the strict "etiquette" of nature (time of day, seasons, and frequency).
- Wants to reach a state of "thoughtlessness" through complex, horizontal compositions.

Pain Points

- Frustrated by beginners who want to play a "hit song" in two days without respecting the "boring" foundation of theory.
- **Dislikes technology that makes music "too easy" (like air-playing or AI)**, as it removes the soul and the struggle of creation.
- Despite his love for practice, his engineering degree creates a "barrier of energy" that makes it hard to maintain his 4-hour daily ritual.
- He focuses on composition and beats; he finds modern lyrics to be a "distraction" from the raw mathematical beauty of the notes.

Empathy Mapping

Empathy Map

HEARS

- If don't know the trendy music then you are boring
- Songs being repeated which they have liked once
- Other people listening to loud music which is umpleasent for them
- "Cheap spa machine" nature tracks or Lo-Fi beats that feel fake and "hollow" compared to real instruments
- People talking at them, office pings, and grammar mistakes in song lyrics that "trigger" their professional brain
- constant honking, the high-pitched whine of rickshaw engines, and the clatter of crowded cafeterias.

SEES

- Trying to uderstand an intrument through youtube tutorial
- People playing amazing instruments and impressiong people .
- When coming back seeing there is noone else so feeling lonely
- Seeing a sudden blast of energy or light when a musical composition builds from silence to full orchestration
- Crowded Metro platforms, the blur of yellow and green rickshaws weaving through traffic, and dusty construction sites.
- A laptop screen filled with "100 open Chrome tabs" lines of code, or ungraded student papers at a time
- An unused guitar or piano under a thin layer of dust—representing a mix of guilt and a "dream self" they don't have time for

SAYS & DOES

- "Nowadays, songs are being made less and remixes are being made more. If you want originality, you have to look for the past."
- "I spend more time skipping than listening because the algorithm just doesn't get my 'vibe' today."
- "Lyrics are a nightmare when I'm tired. My brain starts working again to correct the grammar."
- "I've looked at the piano so many times, but it looks like a giant puzzle. I feel like I'd look foolish trying now."
- "I just want the world to be silent for a second."
- Picks up a guitar or instrument, plays for 3 minutes, feels the pain in the fingers and puts it back —treating it more like a "talisman" than a tool.
- Plugs in earphones the second they leave the office/college to create a "sonic wall" against rickshaws and honking. (60.5% of survey).
- Aggressively clicks the "Next" button while walking or driving, looking for a "feeling" they can't quite name
- Occasionally shuts off the music entirely and sits in total darkness because the search for the "perfect song" became too exhausting. (9.1% of survey).

THINKS & FEELS

- My brain feels like a browser with 50 tabs open. I need a 'Pause Button' before I break. (Mental crowding score of 4/5).
- Why is Spotify playing high-energy pop when I feel grey and heavy? The machine doesn't understand my soul.
- Maybe I'm just not 'musical' I missed my chance as a kid and now my hands are too busy for the struggle
- I'd rather listen to a storm or the ocean than another 'Lo-Fi' beat that sounds like a cheap spa machine. (Reflecting the high "Nature Sounds" appeal in the survey).
- When this note hits, I feel like a firework is going off in a dark room. This sound is a dark blue gradient.
- A feeling of "static" in the head , feel a level 3 to 5 of mental clutter at the end of the day.
- A sense of being misunderstood by their own technology.
- Feeling "bored" or "numb" by short-form music (Reels/TikTok sounds) and craving "Horizontal" long-form peace.
- A deep yearning for "Old is Gold" sounds that feel "thicker" and more meaningful than modern "hollow" pop.

Empathy Map

HEARS

- Searching for that one "thick" sound that drowns out the static of their own thoughts.
- Cringing at the sound of a buzzy string or a wrong note that breaks the "magic bubble.
- The steady hum of a Tanpura or a background piano that gives them a ground to build on.
- Hearing the tiny, extra vibrations that only happen when an instrument is high quality and perfectly in tune.
- That "tick-tick-tick" in the background that feels like a heartbeat or a ticking clock

SEES

- Staring at sheet music, tabs on a screen, or the rhythmic patterns of their own fingers moving.
- Seeing colors (like neon blue or soft green) in their mind when they hit a perfect note or chord.
- Their instrument sitting in a room, looking like a piece of furniture that holds their "true self
- Imagining a "clean room" or a "quiet forest" while they are actually sitting in a messy bedroom.
- Visualizing the way a professional sits or holds their hands to try and copy that "peaceful" look.

SAYS & DOES

- "My fingers are taking a beating, but I wouldn't change it. The pain is how I know I'm learning."
- "I'm not doing this for a concert or for likes on Instagram. I'm doing this to find an anchor in myself."
- "I hate those 'learn guitar in 2 days' apps. It takes a lifetime of mechanical boredom to get to the soul of a song."
- "I wish I had two hours, but I only have fifteen minutes before I'm too tired to hold the instrument."
- "Sometimes I don't even play; I just sit with my instrument and touch the wood. It makes me feel better."
- Spends a long time tuning by ear, refusing to use a phone app because they want to "feel" the note.
- Shifts around constantly, trying to find a seat that doesn't hurt their back or feet while they play.
- Picks up the instrument at midnight or immediately after a stressful commute to "wash off" the day.
- Picks up the instrument at midnight or immediately after a stressful commute to "wash off" the day.

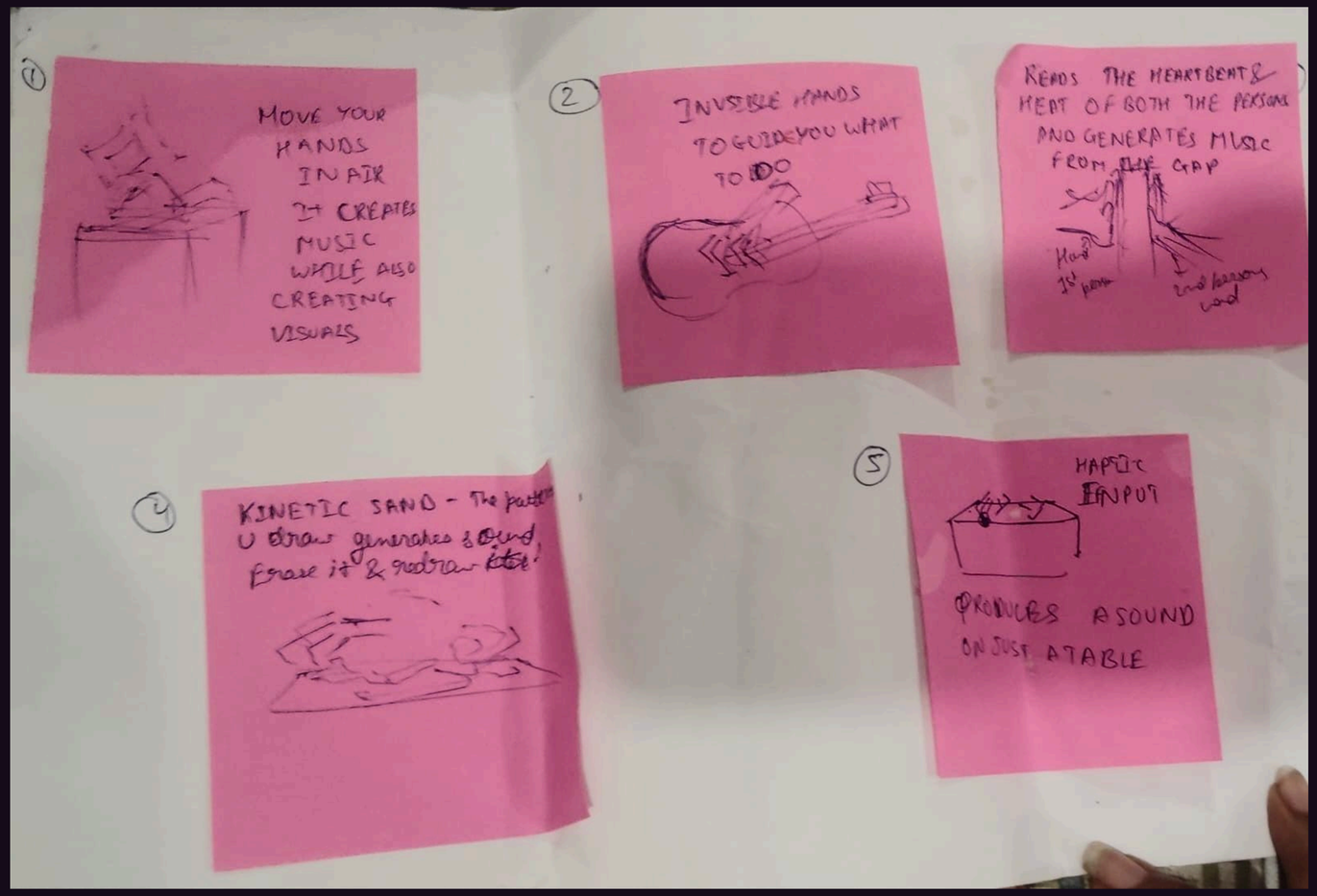
THINKS & FEELS

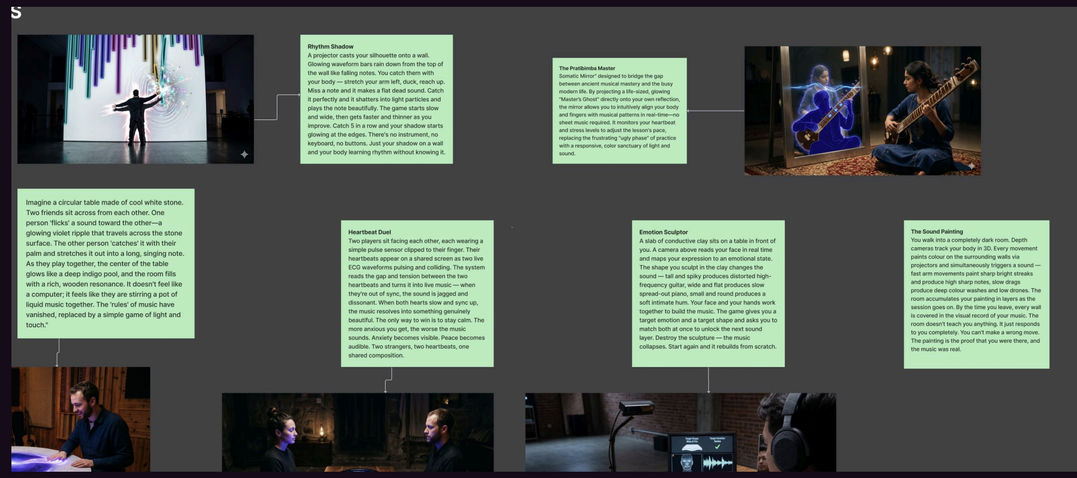
- I wish my brain is as empty as when I was a kid ..Now work keeps me busy
- Is the instrument sounding 'hollow' today, or is it just my mood?
- If I can just get this one part right, the rest of the world will finally go quiet.
- I hope it's not my skill that's the problem—maybe I just need a better quality instrument to reach that 'Deep Blue' feeling.
- Aerospace/Coding is all math, but this Sitar is a different kind of math. It's the math of my heart.
- A mix of frustration and joy when their fingers hurt but the sound is beautiful.
- Losing the track of time while playing music.
- That heavy feeling when they look at their instrument in the corner and realize they've been "too busy" to touch it for a week.
- The feeling of the instrument vibrating against their chest or in their bones, not just their ears.

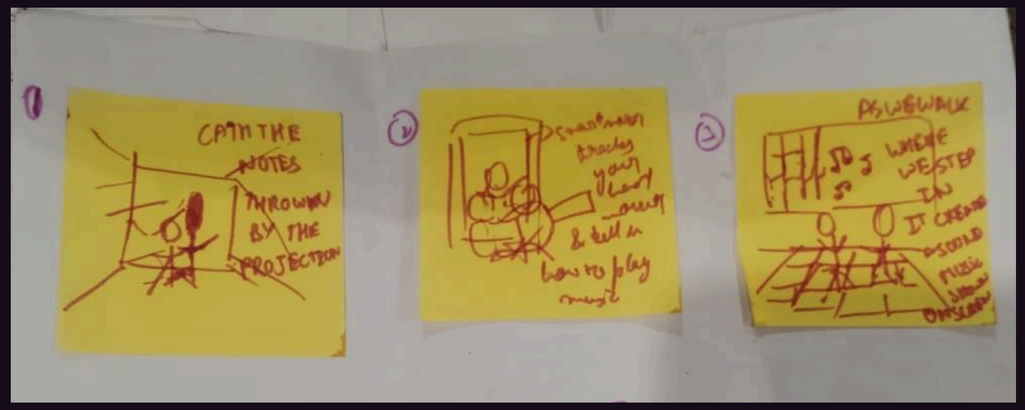
HMW

How might we transform the daily commute for weary listeners so that their music experience gives them energy instead of just filling the silence?

INDIVIDUAL BRAINSTORMING

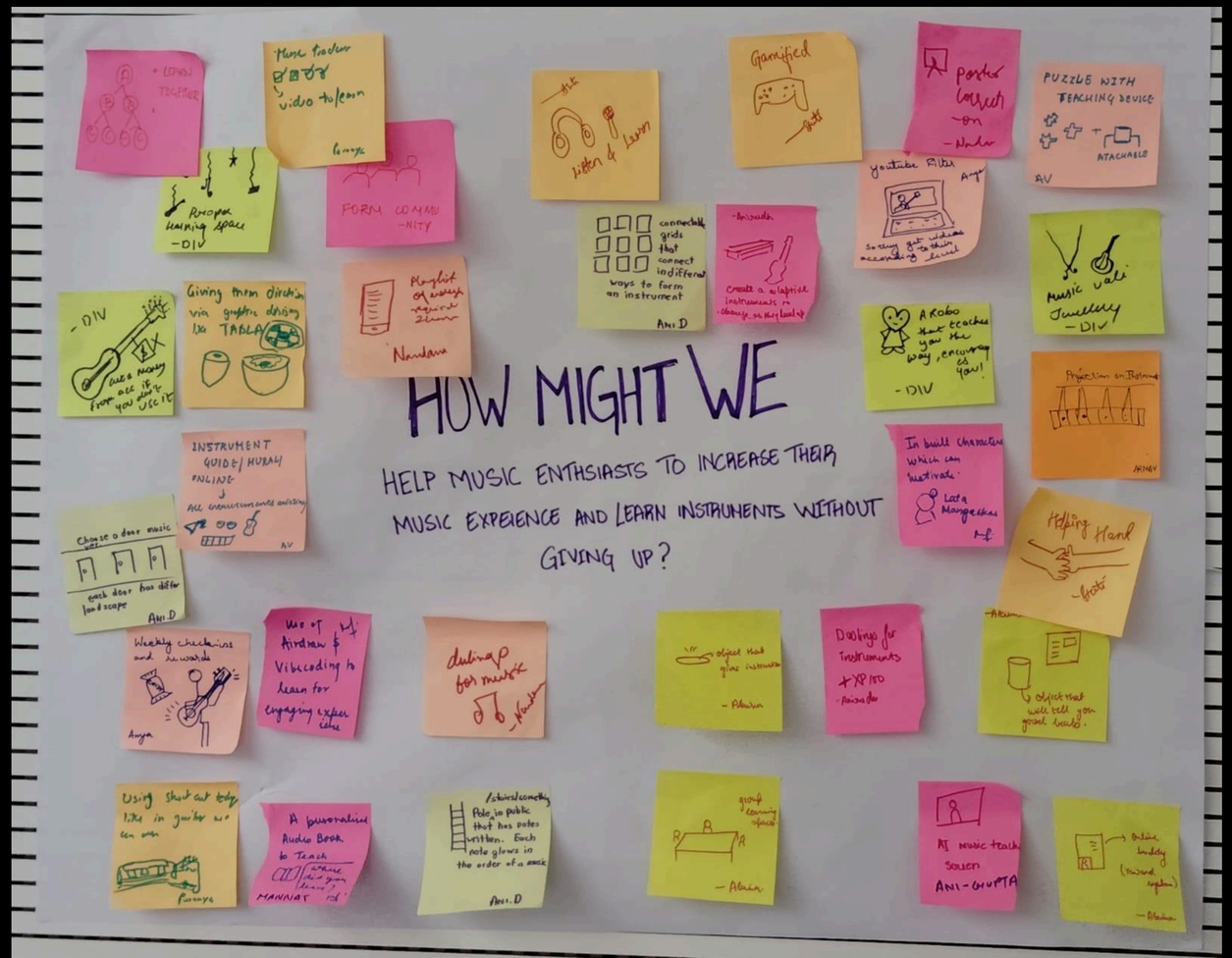






DATE

GROUP BRAINSTORMING



CAPTION

FINAL CONCEPTS

Concepts 1

Rhythm Shadow

A projector casts your silhouette onto a wall.

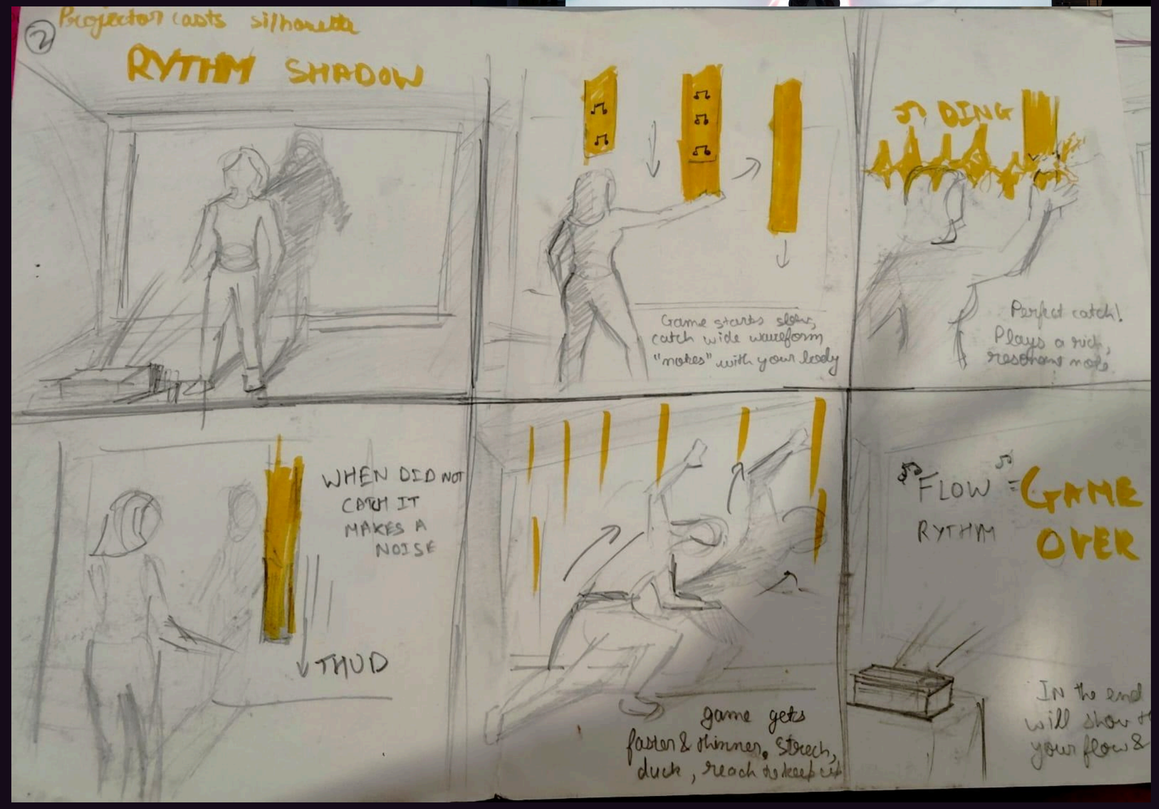
Glowing waveform bars rain down from the top of the wall like falling notes.

You catch them with your body — stretch your arm left, duck, reach up. Miss a note and it makes a flat dead sound.

Catch it perfectly and it shatters into light particles and plays the note beautifully.

The game starts slow and wide, then gets faster and thinner as you improve. Catch 5 in a row and your shadow starts glowing at the edges.

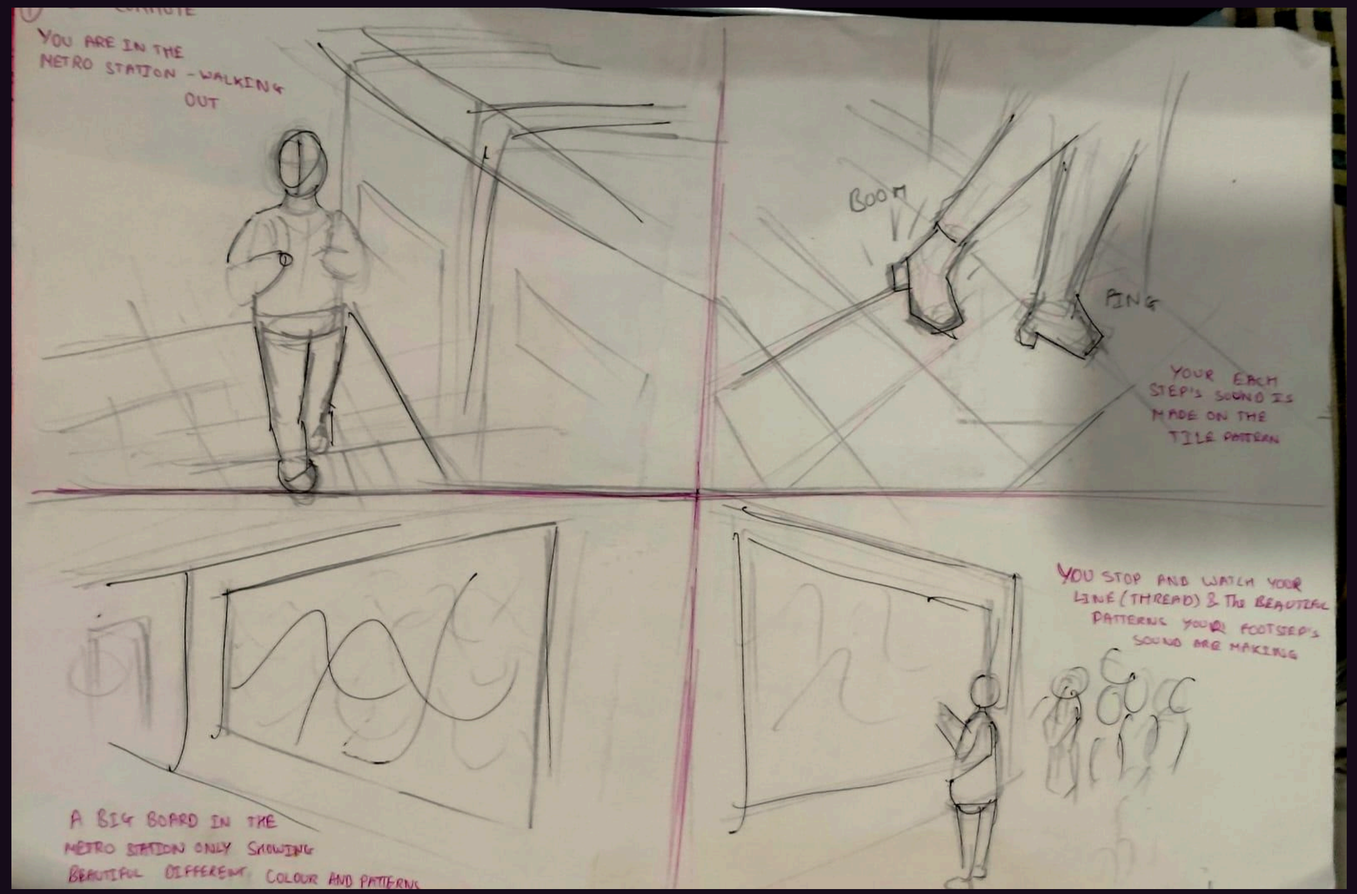
There's no instrument, no keyboard, no buttons. Just your shadow on a wall and your body learning rhythm without knowing it.



Concepts 2

City Orchestra

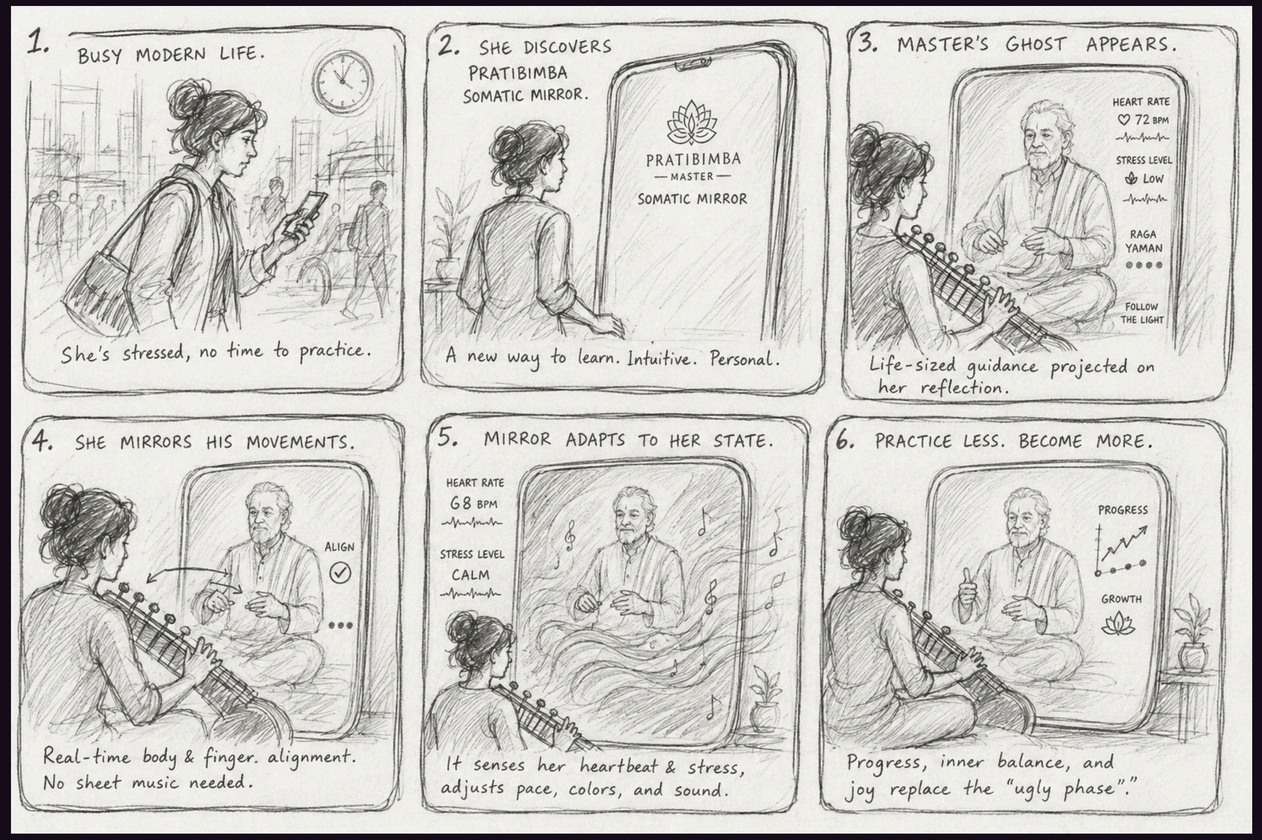
Pressure-sensitive tiles embedded in a metro station floor. Every commuter's footstep triggers a note — heavy steps produce low notes, light steps produce high ones, faster walking speeds up the tempo. A massive screen on the platform wall shows a live visualisation — each person is a unique coloured thread, their movement weaving into a generative orchestral tapestry in real time. Nobody signed up to play. Nobody knows they're part of it at first. Then someone stops and looks at the screen and realises their thread is woven into that music. The rush hour crowd becomes an accidental orchestra. The daily commute that drains people of energy accidentally gives it back to them.



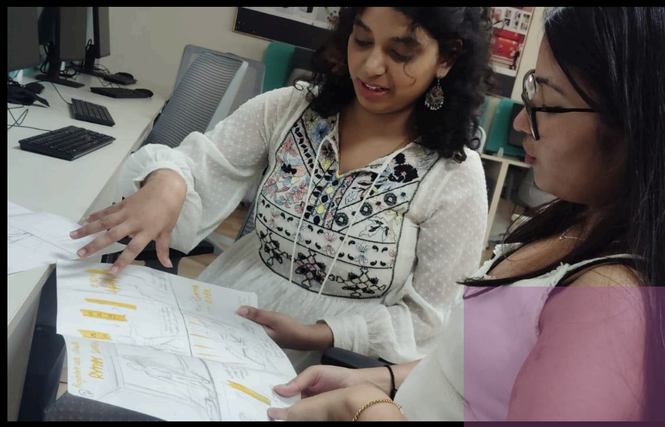
Concepts 3

The Pratibimba Master

Somatic Mirror" designed to bridge the gap between ancient musical mastery and the busy modern life. By projecting a life-sized, glowing "Master’s Ghost" directly onto your own reflection, the mirror allows you to intuitively align your body and fingers with musical patterns in real-time—no sheet music required. It monitors your heartbeat and stress levels to adjust the lesson's pace, replacing the frustrating "ugly phase" of practice with a responsive, color sanctuary of light and sound.



Concept Testing



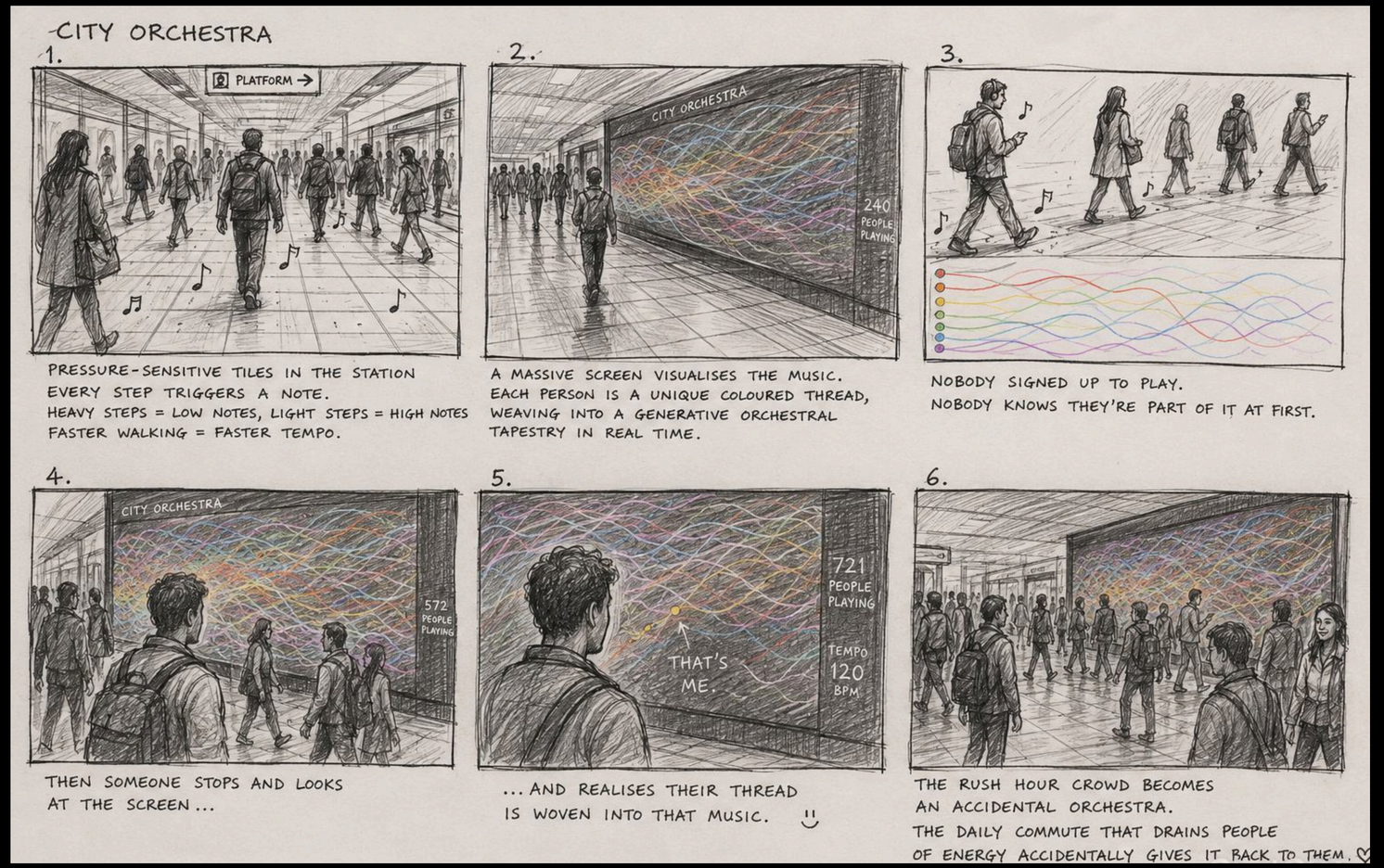
CITY ORCHESTRA

Converting Urban Stress into Generative Harmony.

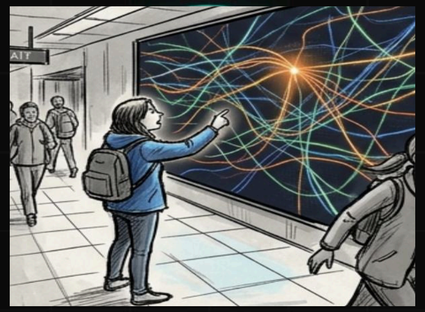
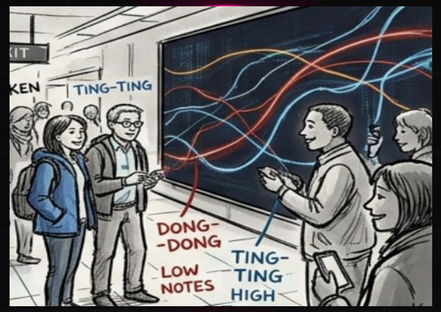
The Core Concept

1. **A Commuter** walks across the platform
2. **Every Step: Triggers a musical note**
3. Heavy steps create low, resonant notes. Fast walking translates to high tempo rhythm.
4. Action: An overhead webcam (mounted high on a tripod) tracks the commuter's position.
5. The software matches the "Sound" from Step 3 to the "Coordinate" of the person on the floor.

The Result: A digital "thread" is drawn on the screen exactly where the person is standing. As they walk, the thread follows them, "weaving" through the threads of other commuters.



User Journey Map

ENTRY	WALKING	DISCOVERY	PLAY	EXIT
				

😩 EXHAUSTED

Action
Pushing through the dense crowd.

Pain Point
High city noise increasing heart rate and anxiety.

🤨 CONFUSED

Action
Hears a rhythmic pulse in sync with footsteps.

Touchpoint
Pressure-sensitive floor tiles.

😲 SURPRISED

Action
Realizes screen tracks their movement.

Touchpoint
The "Master's Ghost" / Personal Mandala.

😊 ENERGIZED

Action
Intentionally walking to "weave" music.

Transformation
"Shield" of music replaced by "Expression."

😌 SHANTI

Action
Boards train with regulated nervous system.

Outcome
Commute gave energy back instead of draining

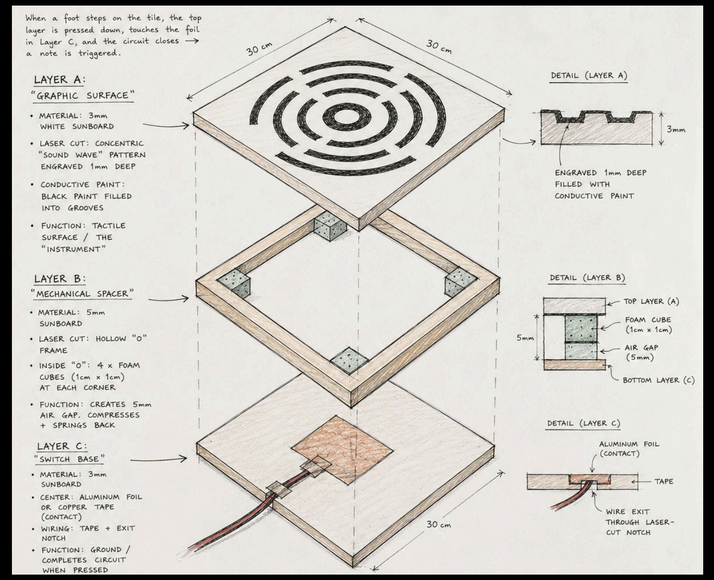
The Making Process

The Eye: Webcam tracking movement.

The Human: The commuter acting as the performer.

The Interface: Laser-cut sunboard tiles with conductive paint.

The Brain: Laptop & Arduino processing the data.



Multimodal Framework

Auditory

Generative orchestral feedback (Low/High notes).

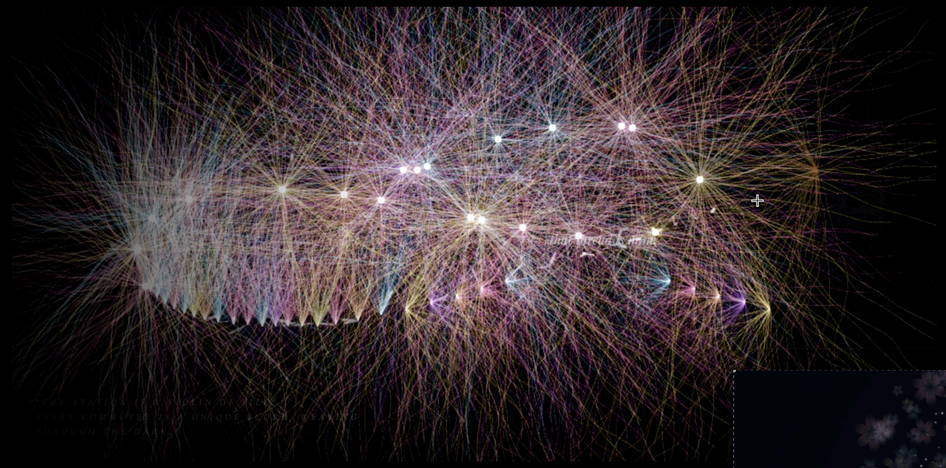
Visual

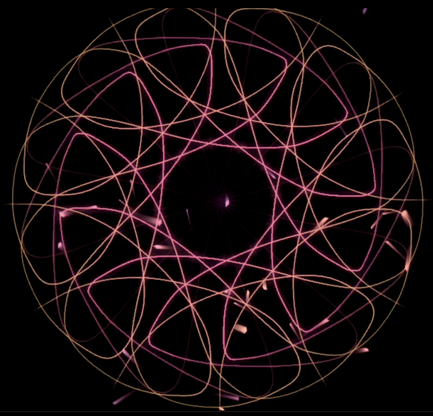
The "Mandala" threads weaving in real-time , tracked by webcam

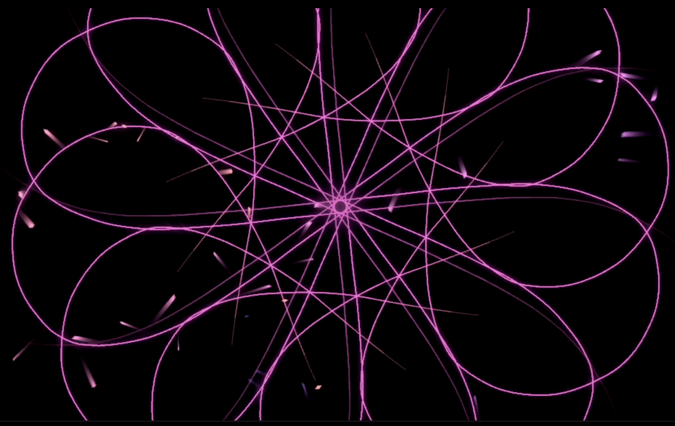
Haptic

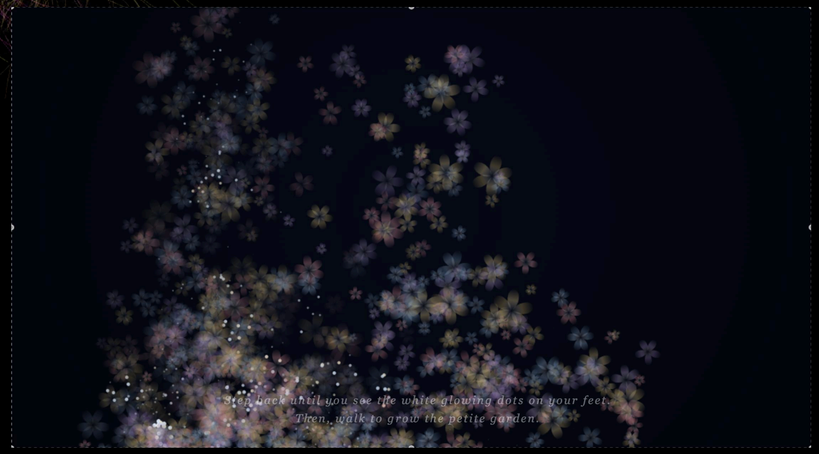
The physical "spring-back" resistance of the foam-layered tiles.

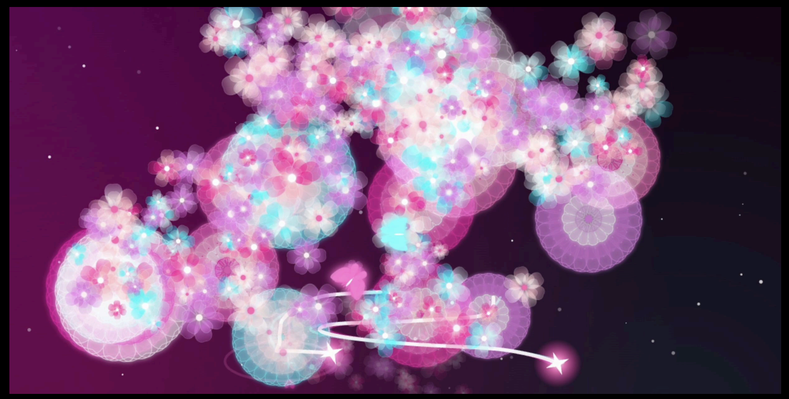
Tested all the visulization that can happen







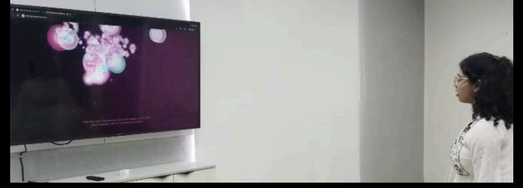


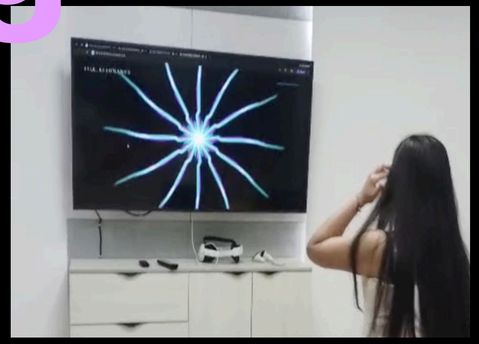


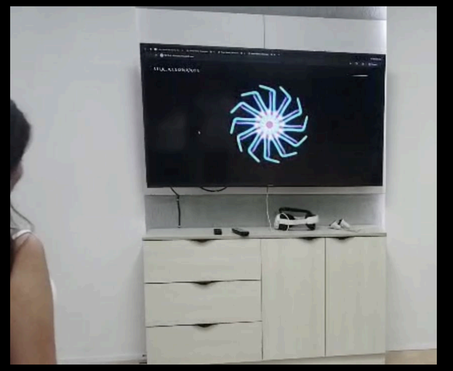
these are all the visualization made while tracking feet

User Testing

Tested different different visualization with people and understood that they liked manadala one more







Link to the testing
https://drive.google.com/drive/folders/1fNmA1caRR3UxZx4qZVkaHVmWFWpYIcaQ?usp=share_link

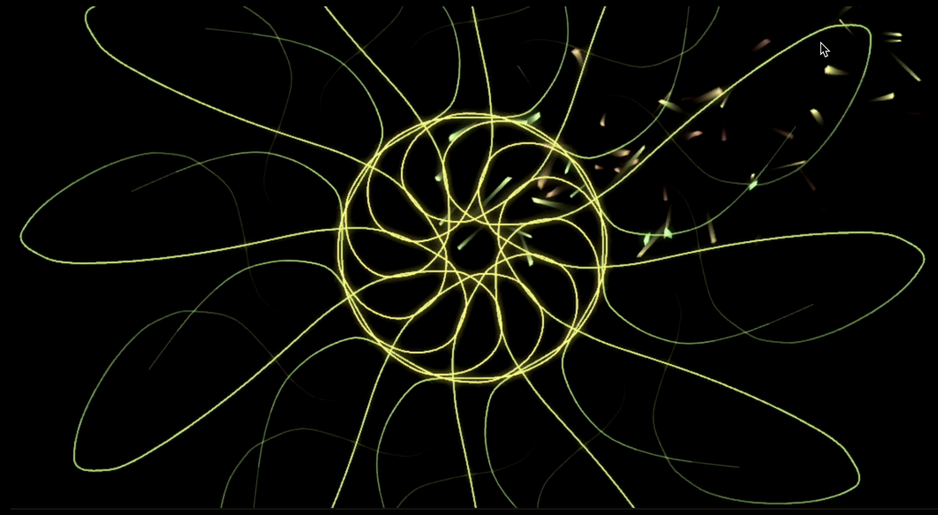
User Testing Results:

Aesthetic Preference Score

The Mandala (Geometric): 75% Approval. (The clear winner). Users felt it provided a sense of "Shanti" (order and peace) that matched the Indian Classical logic.

The Flowers (Organic): 15% Approval. (Strong gender bias). 80% of female testers liked it, but 90% of male testers found it "too soft."

The Strings (Original Concept): 10% Approval. Rated as "visually noisy" and "anxiety-inducing" in a crowded rush-hour context.



Key Findings:

1: The Mandala was the most successful UI because it balanced **"visual complexity" with "mental calm."**

2: Strings and Hybrid themes failed due to "Cognitive Overload." Users in a busy metro don't want more "mess"—they want "order."

Technical Finding: Feet detection accuracy dropped by 33% when users ran. This defined our system limit: "Designed for the Stroller, not the Sprinter."

Iterative Change: Replaced "Thin Threads" with "Bold Flower Mandalas" to improve visibility and tracking stability

Conclusion

Outcome 1: Successfully transformed a passive urban space into an active creative sanctuary.

Outcome 2: Lowered the entry barrier to music—0 hours of practice required to compose.

Rationale: Design isn't just about utility; it's about reclaiming peace in the machine.
drive link-https://drive.google.com/drive/folders/1fNmA1caRR3UxZx4qZVkaHVmWFWpYIcaQ?usp=share_link